

Andrew DeLaCruz

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EDUCATION

University of California, Davis

Davis, CA

B.S. Mechanical Engineering, Sophomore | GPA: 3.30/4.00

Expected Graduation: Jun 2029

- **Completed:** Engineering Programming (MATLAB)
- **In Progress:** Statics, Dynamics, Thermodynamics, Engineering Graphics, Circuits I & II, Properties of Materials

EXPERIENCE

Cogwell, Inc. Santa Clara, CA

Jun 2025 – Sep 2025

Calibration & Test Engineering Intern

Jun 2026 – Sep 2026

- Automated calibration of 9 instrument models (Keysight, Fluke, Tektronix) by writing Fluke MET/CAL procedures and Python scripts, producing calibration certificates up to 20x faster and more reliably
- Diagnosed circuit-level PCB faults through schematic analysis and repaired boards with solder rework, returning failed instruments to service
- Validated 30+ precision instruments against calibration standards and technical specifications using multimeters, oscilloscopes, spectrum analyzers, and signal generators, confirming each met spec before return to customers

PROJECTS

Nightstand Hub | SolidWorks, GD&T, FDM 3D Printing, ESP32, Embedded Systems

Aug 2026 – Present

- Designed a three-part, 11 × 8 × 16 cm SolidWorks enclosure packaging two displays, an ESP32, a speaker, and a temperature sensor, dimensioned with calipers and micrometers; delivered a prototype for \$31.72
- Designed press fits and locating pegs that eliminated adhesives and mounting brackets, reaching a working fit in 2 print iterations (DFM); documented critical fits in GD&T drawings using datums
- Isolated the temperature sensor from heat-generating electronics with a tapered top and rear-panel airflow path, preventing self-heating error; verified readings against an external thermometer
- Traced a full display failure during system bring-up to a mis-held GPIO pin through pull-up/pull-down testing; rewired each display and sensor onto a dedicated bus, eliminating a recurring intermittent fault
- Integrated weather, calendar (iCal), and transit APIs into ESP32 firmware through AI-assisted development (Claude Code), validating each subsystem on hardware; the device is now in daily use

FlashBack: 270° Panoramic Photo Booth | SolidWorks, Python, OpenCV, Raspberry Pi

Mar 2026 – May 2026

- Co-designed a battery-powered rotating photo booth with a 22-person team in 6 weeks, delivering a field-tested \$339.60 prototype for DUEN events; served on the EECS team and as mechanical liaison
- Moved chassis load off the servo shaft with a ball-bearing base and added counterweights, eliminating motor-induced rattle and keeping the booth stable with the mast extended
- Packaged a Raspberry Pi 4, battery, buck converters, fuse box, and thermal printer with planned cable routing; sized the power budget for 6-hour battery runtime at events
- Fixed unreliable panorama stitching by tuning image overlap, exposure locking, and PWM servo settle time in a Python/OpenCV pipeline, making camera rotation 3x faster while raising stitch success to up to 90%
- Traced a Raspberry Pi short circuit minutes before technical review and swapped in a backup, keeping the demo on schedule; assembled the booth using heat-set inserts and soldered wiring

LEADERSHIP

Davis Undergraduate Engineering Network (DUEN)

Mar 2026 – Present

Officer, Finance (May 2026 – Present); Engineering Intern (Mar – May 2026)

Davis, CA

- Manage club finances, tracking member dues and project spending in Excel to keep the budget accurate
- Plan restaurant fundraisers with local Davis businesses and coordinate with officers to fund project builds
- Interview new applicants and mentor new members through their hands-on build projects

TECHNICAL SKILLS

CAD & Prototyping: SolidWorks, GD&T, engineering drawings, DFM, FDM 3D printing, heat-set inserts, soldering

Hardware & Test: Embedded systems (ESP32, Raspberry Pi), oscilloscopes, multimeters, spectrum analyzers

Software: Python, MATLAB, Fluke MET/CAL, OpenCV, Git, Claude Code

Languages: English, Spanish